OPPERMANN INTERVALS IN $\mathbb{F}_q[t]$: COUNTEREXAMPLES IN CHARACTERISTIC 2 AND A COMPUTER-ASSISTED THEOREM IN ODD CHARACTERISTIC

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ABSTRACT. For a monic $f \in \mathbb{F}_q[t]$ of degree d the Legendre interval $\{f^2 + s : \deg s \le d\}$ splits into the q *Oppermann sub-intervals* $O_{f,c} = \{f^2 + cf + s : \deg s < d\}$, $c \in \mathbb{F}_q$, each of size q^d; for $q = 2$ these are the two halves, the analogue of (n^2, n^2+n) and $(n^2+n, (n+1)^2)$. We ask whether every $O_{f,c}$ contains an irreducible polynomial. In characteristic 2 the answer is no, and the failures are explained completely by three Galois-theoretic obstructions: (A) for $d = 1$ the piece $c = 0$ consists of squares; (B) for $d = 2$ we compute the exact number of irreducibles in every Oppermann sub-interval, $q^2/4$, 0, $q(q-2)/4$ or $q(q-1)/2$, by letting Frobenius act on the roots of an additive polynomial through $\mathrm{AGL}_2(\mathbb{F}_2) \cong S_4$; it vanishes exactly when $q = 4^j$, $a_1^2 + c = 0$, or $q = 2$, $a_1^2 + c = 1$; (C) for $q = 2$, $d = 4$ the block $\{t^8 + s : \deg s \le 3\}$ contains no irreducible, by the Stickelberger–Swan sign of Frobenius, because $8 \mid \deg$ makes the sign depend only on the tail. In odd characteristic every class of the Hayes group of level d is an Oppermann sub-interval, and an exact computation with the Hayes explicit formula shows that all of them contain irreducibles for $q = 3$, $d \le 16$; $q = 5$, $d \le 10$; $q = 7$, $d \le 9$; $q = 9$, $d \le 8$; $q = 11$, $d \le 7$; $q = 13$, $d \le 6$. In characteristic 2 the same computation, for $q \in \{2, 4, 8, 16\}$ up to $q^d \le 2^{26}$, finds no empty Oppermann sub-interval beyond (A)–(C). The Weil bound sits exactly at the square-root barrier for these intervals and cannot prove nonemptiness for any $d \ge 2$.

1. INTRODUCTION

Oppermann's conjecture [6] asserts that for every $n \ge 2$ both (n^2, n^2+n) and $(n^2+n, (n+1)^2)$ contain a prime. It implies Legendre's conjecture and lies beyond the Riemann hypothesis, which gives primes only in intervals $[x, x + C\sqrt{x}\log x]$. Over $\mathbb{F}_q[t]$, Legendre intervals $I_f = \{f^2 + s : \deg s \le d\}$ (f monic of degree d) were studied in [3]. Since $f^2 + cf$ and $f^2 + ct^d$ differ by a polynomial of degree $< d$, the interval I_f is the disjoint union of the q sub-intervals

$$O_{f,c} = \{f^2 + cf + s : \deg s < d\} = \{f^2 + ct^d + s : \deg s < d\}, \qquad c \in \mathbb{F}_q, \tag{1}$$

each of cardinality $q^d = |f|$, the norm of f, exactly as $(n^2, n^2 + n)$ has length n. For $q = 2$ there are two halves. We call $O_{f,c}$ the *Oppermann sub-intervals* and ask:

Does every $O_{f,c}$ contain an irreducible polynomial?

The answer depends on the characteristic.

Theorem 1.1 (characteristic 2)**.** *Let $q = 2^k$.*

(A) *For $d = 1$ the q sub-intervals $O_{f,0}$ consist of squares.*

(B) *For $d = 2$ and $f = t^2 + a_1 t + a_0$, the number of irreducibles in $O_{f,c}$ depends only on $e = a_1^2 + c$ and equals*

	$e = 0$	$e \neq 0$
k *even*	0	$q^2/4$
k *odd*	$q(q-1)/2$	$q(q-2)/4$

In particular $O_{f,c}$ is empty if and only if either k is even and $c = a_1^2$, or $q = 2$ and $c = a_1^2 + 1$; this gives q^2 empty sub-intervals for every $q \in \{2, 4, 16, 64, \dots\}$ and none for $q \in \{8, 32, 128, \dots\}$.

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(C) *For $q = 2$ and $d = 4$ the block $\{t^8 + s : \deg s \le 3\}$ contains no irreducible polynomial; it is $O_{f,c}$ for the 8 pairs $f = t^4 + a_2t^2 + a_1t + a_0$, $c = a_2$.*

Theorem 1.2 (odd characteristic, computer-assisted)**.** *Let q be odd. Every Oppermann sub-interval $O_{f,c}$ contains an irreducible polynomial whenever*

$$q = 3,\ d \le 16; \qquad q = 5,\ d \le 10; \qquad q = 7,\ d \le 9;$$
$$q = 9,\ d \le 8; \qquad q = 11,\ d \le 7; \qquad q = 13,\ d \le 6.$$

Equivalently (Lemma 2.1), every interval $\{P + s : \deg s < d\}$ with P monic of degree $2d$ contains an irreducible polynomial in these ranges.

Theorem 1.3 (characteristic 2, computer-assisted)**.** *For $q = 2$, $d \le 26$; $q = 4$, $d \le 13$; $q = 8$, $d \le 7$; $q = 16$, $d \le 5$, the only empty Oppermann sub-intervals are those of Theorem 1.1.*

Section 2 identifies the Oppermann sub-intervals with Hayes classes and explains why characteristic 2 is special: the Oppermann classes are concentrated on prefixes close to squares. Section 3 proves Theorem 1.1. Section 4 describes the exact computation behind Theorems 1.2 and 1.3. Section 5 explains why the explicit formula alone cannot go further.

2. Oppermann sub-intervals as Hayes classes

For P monic of degree n write $P = t^n + p_{n-1}t^{n-1} + \cdots + p_0$. Two monic polynomials of degree n lie in the same *Hayes class of level l* if they agree in $p_{n-1}, \dots, p_{n-l}$; the classes are the cosets of a group G_l of order q^l (the series $1 + p_{n-1}u + \cdots + p_{n-l}u^l \bmod u^{l+1}$ under multiplication), and their characters are the short-interval characters of Hayes [4] and Keating–Rudnick [5]. By (1), $O_{f,c}$ is the Hayes class of level d and degree $n = 2d$ containing $f^2 + ct^d$.

Lemma 2.1 (odd q)**.** *If q is odd, every Hayes class of level d in degree $2d$ is an Oppermann sub-interval, and it equals $O_{f,c}$ for exactly q pairs (f, c).*

Proof. Write $f = t^d + a_{d-1}t^{d-1} + \cdots + a_0$. The coefficient of t^{2d-j} in f^2 ($1 \le j \le d$) is $2a_{d-j} + \Phi_j(a_{d-1}, \dots, a_{d-j+1})$. Since 2 is invertible, the coefficients of $t^{2d-1}, \dots, t^{d+1}$ determine $a_{d-1}, \dots, a_1$ uniquely, and the coefficient of t^d in $f^2 + ct^d$ is $2a_0 + \Phi_d + c$, which takes a prescribed value for exactly q pairs (a_0, c). □

So in odd characteristic the Oppermann question at degree $2d$ is the question whether *every* short interval of length $q^d = q^{n/2}$ contains a prime: no class is singled out.

Lemma 2.2 (q even)**.** *If q is even, $O_{f,c}$ is the class whose prefix has coefficient 0 at every odd j with $d < j < 2d$ and arbitrary coefficients at $j = d$ and at the even j with $d < j < 2d$. There are $q^{\lceil d/2 \rceil}$ such classes among the q^d classes of level d.*

Proof. In characteristic 2, $f^2 = \sum a_i^2 t^{2i}$ has no odd terms, and $a \mapsto a^2$ is bijective on $\mathbb{F}_q$; the coefficient at t^d is shifted by c. □

Thus in characteristic 2 the Oppermann sub-intervals are the short intervals around the $q^{\lceil d/2 \rceil}$ "near-squares" $f^2 + ct^d$, a set of measure $q^{-\lfloor d/2 \rfloor}$ among all short intervals. Squares and near-squares carry Galois-theoretic structure, and that structure produces the counterexamples below.

3. Characteristic 2: the three obstructions

Throughout this section $q = 2^k$ and $\mathrm{Tr} = \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}$.

3.1. **(A)** $d = 1$**.** $O_{f,0} = \{f^2 + s_0 : s_0 \in \mathbb{F}_q\} = \{(f + \sqrt{s_0})^2\}$ consists of squares.

3.2. **(B) $d = 2$: an additive polynomial and $\mathrm{AGL}_2(\mathbb{F}_2)$.** For $f = t^2 + a_1 t + a_0$ we have $f^2 = t^4 + a_1^2 t^2 + a_0^2$, so

$$O_{f,c} = B_e := \{t^4 + et^2 + \alpha t + \beta : \ \alpha, \beta \in \mathbb{F}_q\}, \qquad e = a_1^2 + c.$$

By Lemma 2.2 the blocks B_e, $e \in \mathbb{F}_q$, are precisely the Oppermann classes for $d = 2$.

Proposition 3.1. *Let $\alpha \neq 0$. Put $L(t) = t^4 + et^2 + \alpha t$ and $R(y) = y^3 + ey + \alpha$.*

1. *$t^4 + et^2 + \alpha t + \beta$ is irreducible for some β only if R has exactly one root in $\mathbb{F}_q$.*
2. *If R has exactly one root in $\mathbb{F}_q$, then $t^4 + et^2 + \alpha t + \beta$ is irreducible for exactly $q/2$ values of $\beta \in \mathbb{F}_q$.*

If $\alpha = 0$, every member $t^4 + et^2 + \beta = (t^2 + \sqrt{e}\, t + \sqrt{\beta})^2$ is a square.

Proof. L is $\mathbb{F}_2$-linear and separable ($L' = \alpha \neq 0$), so $V = \ker L \subset \overline{\mathbb{F}}_q$ is a 2-dimensional $\mathbb{F}_2$-space, and $V \setminus \{0\}$ is the set of roots of $L(t)/t = R(t)$. The Frobenius $\varphi(x) = x^q$ acts on V by some $F \in \mathrm{GL}(V) \cong S_3$. Exactly one of three cases holds: $F = 1$ (R has three roots in $\mathbb{F}_q$), F of order 3 (no root), or F a transvection (exactly one root).

Fix β and a root r of $P_\beta = L + \beta$. The roots of P_β form the coset $r + V$. Since $L(\varphi(r)) = \varphi(\beta) = \beta$, we have $\varphi(r) = r + v$ with $v \in V$, and $\varphi(r + w) = r + (Fw + v)$. So Frobenius acts on the roots through the affine map $T(w) = Fw + v$ of $V \cong \mathbb{F}_2^2$, an element of $\mathrm{AGL}_2(\mathbb{F}_2) \cong S_4$. P_β is irreducible if and only if T is a 4-cycle.

- If $F = 1$, T is a translation and $T^2 = 1$.
- If F has order 3, $F - 1$ is invertible, so T has a fixed point.
- If F is a transvection, $T^2(w) = w + (F+1)v$, so T is a 4-cycle if and only if $(F+1)v \neq 0$, i.e. $v \notin V^F$.

This proves (1). For (2), let F be a transvection, so $(F + 1)V = V^F$ is a line. Replacing r by $r + w$ replaces v by $v + (F + 1)w$, so the class $[v] \in V/V^F \cong \mathbb{F}_2$ depends only on β, and $\beta \mapsto [v]$ is additive. Its kernel consists of the β for which some root $r' = r + w$ is fixed by φ, i.e. $\beta \in L(\mathbb{F}_q)$. The map $L : \mathbb{F}_q \to \mathbb{F}_q$ is $\mathbb{F}_2$-linear with kernel $V^F = V \cap \mathbb{F}_q$ of order 2, so $|L(\mathbb{F}_q)| = q/2$ and exactly $q/2$ values of β have $[v] \neq 0$. □

Lemma 3.2. *Let $\alpha \neq 0$. The cubic $R(y) = y^3 + ey + \alpha$ has exactly one root in $\mathbb{F}_q$ if and only if $\mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)$.*

Proof. The roots y_1, y_2, y_3 of R are distinct (R is separable as above), with $y_1 + y_2 + y_3 = 0$ and $y_1 y_2 y_3 = \alpha$. Put $\rho = y_1/y_3 + y_2/y_1 + y_3/y_2$ and $\rho' = y_2/y_3 + y_1/y_2 + y_3/y_1$. Even permutations of the roots fix ρ and odd ones exchange ρ, ρ'; using $y_i + y_j = y_k$,

$$\rho + \rho' = \frac{y_1 + y_2}{y_3} + \frac{y_2 + y_3}{y_1} + \frac{y_3 + y_1}{y_2} = 1.$$

Hence ρ, ρ' are the roots of $z^2 + z + \delta$ with $\delta = \rho\rho' \in \mathbb{F}_q$, and Frobenius acts on the roots of R by an odd permutation (a transposition: exactly one root in $\mathbb{F}_q$) if and only if $\rho \notin \mathbb{F}_q$, i.e. $\mathrm{Tr}(\delta) = 1$. Expanding,

$$\delta = 3 + \sum_k \frac{y_i y_j}{y_k^2} + \sum_k \frac{y_k^2}{y_i y_j} = 1 + \sum_k \frac{\alpha}{y_k^3} + \frac{1}{\alpha} \sum_k y_k^3.$$

From $y_k^3 = ey_k + \alpha$ we get $\sum_k y_k^3 = \alpha$, so $\delta = \sum_k \alpha/w_k$ with $w_k = y_k^3 = ey_k + \alpha$. Substituting $y = (w + \alpha)/e$ (for $e \neq 0$) into $w = y^3$ shows that the w_k are the roots of $w^3 + \alpha w^2 + (\alpha^2 + e^3)w + \alpha^3$, so $\sum_k 1/w_k = (\alpha^2 + e^3)/\alpha^3$ and

$$\delta = 1 + e^3/\alpha^2.$$

For $e = 0$ this also holds directly ($w_k = \alpha$). So $\mathrm{Tr}(\delta) = 1$ if and only if $\mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)$. □

Remark 3.3. δ is Berlekamp's characteristic-2 discriminant [2] of R, and R is the cubic resolvent of the quartic; the proof above is self-contained. For $e = 0$ the lemma says: $y^3 = \alpha$ has exactly one root iff k is odd, which is the familiar fact that cubing is bijective on $\mathbb{F}_q$ iff $3 \nmid q - 1$.

Proof of Theorem 1.1(B). By Proposition 3.1, $|B_e \cap \{\text{irreducibles}\}| = \frac{q}{2} \cdot \#\{\alpha \neq 0 : \mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)\}$. If $e = 0$ the condition reads $\mathrm{Tr}(1) = 1$, i.e. k odd: the count is $\frac{q}{2}(q-1)$ for k odd and 0 for k even. If $e \neq 0$, $\alpha \mapsto e^3/\alpha^2$ permutes $\mathbb{F}_q^\times$, and $\#\{x \neq 0 : \mathrm{Tr}(x) = 1\} = q/2$, $\#\{x \neq 0 : \mathrm{Tr}(x) = 0\} = q/2 - 1$; with $\mathrm{Tr}(1) = k \bmod 2$ this gives $q^2/4$ (k even) and $\frac{q}{2}(\frac{q}{2} - 1)$ (k odd). The count $q(q-2)/4$ vanishes only for $q = 2$. □

The formula was checked by brute force for $q \leq 64$ (script `char2_d2_theorem.py`): criterion of Lemma 3.2, the $q/2$ count of Proposition 3.1(2), and the four block totals.

3.3. **(C) $q = 2$, $d = 4$: the sign of Frobenius.** The Oppermann classes for $q = 2$, $d = 4$ are the four prefixes $(p_7, p_6, p_5, p_4) = (0, x, 0, y)$ (Lemma 2.2). For $x = y = 0$ the class is the *zero block*

$$Z_4 = \{t^8 + a_3t^3 + a_2t^2 + a_1t + a_0\} \subset \mathbb{F}_2[t],$$

equal to $O_{f,c}$ for $f = t^4 + a_2t^2 + a_1t + a_0$ and $c = a_2$.

We use the theorem of Stickelberger [7] in the form of Swan [8, Cor. 1]: *if $\tilde{P} \in \mathbb{Z}[t]$ is monic of degree n and its reduction $P \in \mathbb{F}_2[t]$ is squarefree with r irreducible factors, then $r \equiv n \pmod 2$ if and only if* $\mathrm{disc}(\tilde{P}) \equiv 1 \pmod 8$. Equivalently, the Frobenius permutation of the roots of P is even if and only if $\mathrm{disc}(\tilde{P}) \equiv 1 \pmod 8$. An irreducible P of degree 8 has Frobenius an 8-cycle, an odd permutation, and would need $\mathrm{disc}(\tilde{P}) \equiv 5 \pmod 8$.

Lemma 3.4. *Let $\tilde{P} = t^n + \tilde{g}$ with $\tilde{g} \in \mathbb{Z}[t]$, $\deg \tilde{g} < n$, and $8 \mid n$. Then* $\mathrm{disc}(\tilde{P}) \equiv \mathrm{Res}(\tilde{P}, \tilde{g}') \pmod 8$.

Proof. $\mathrm{disc}(\tilde{P}) = (-1)^{n(n-1)/2} \mathrm{Res}(\tilde{P}, \tilde{P}') = \prod_{\tilde{P}(\rho)=0}(n\rho^{n-1} + \tilde{g}'(\rho))$, the sign being $+1$ for $8 \mid n$. Expanding the product, every term containing a factor n is n times a symmetric polynomial with integer coefficients in the roots, hence lies in $8\mathbb{Z}$. □

Proof of Theorem 1.1(C). An irreducible member of Z_4 has no root in $\mathbb{F}_2$, so $a_0 = 1$ and $a_1 + a_2 + a_3 = 1$. This leaves four polynomials. $t^8 + t^2 + 1 = (t^4 + t + 1)^2$ is a square. For the other three choose integer lifts and apply Lemma 3.4:

- $t^8 + t + 1$: $\tilde{g}' = 1$, disc $\equiv 1$;
- $t^8 + t^3 + 1$: $\tilde{g}' = 3t^2$, $\mathrm{Res}(\tilde{P}, 3t^2) = 3^8\big(\prod \rho\big)^2 = 3^8 \equiv 1$;
- $t^8 + t^3 + t^2 + t + 1$, lifted as $t^8 + (t+1)^3$: $\tilde{g}' = 3(t+1)^2$, and $\mathrm{Res}(\tilde{P}, 3(t+1)^2) = 3^8\tilde{P}(-1)^2 = 3^8 \equiv 1$.

In each case disc $\equiv 1 \pmod 8$, so the number of irreducible factors is even. □

Remark 3.5. The mechanism is that $8 \mid \deg P$ kills the leading term of P' modulo 8, so the sign of Frobenius is decided by the tail g alone; with $\deg g \leq 3$ the tail is too short to produce the sign -1 on a polynomial with no linear factor. Table 1 shows the zero blocks $Z_d = \{t^{2d} + s : \deg s < d\}$, which are Oppermann classes for every d. For $d = 8$ ($8 \mid 2d$ again) the tail has degree ≤ 7, half of the squarefree members already have the sign -1, and the block contains 10 irreducibles. The parallel with (B) is exact: in (B) the obstruction is the quotient $S_4 \to S_3$ (the resolvent cubic), in (C) the quotient $S_8 \to \{\pm 1\}$ (the sign).

4. The exact computation

4.1. **Method.** For a class x of level l and degree n, let $\Psi(x) = \sum_{F \in x} \Lambda(F)$. The explicit formula [4, 5] reads

$$\Psi(x) = q^{n-l} + q^{-l} \sum_{\chi \neq 1} \overline{\chi}(x)\, \psi_n(\chi), \qquad u\frac{L'(u,\chi)}{L(u,\chi)} = \sum_{m \geq 1} \psi_m(\chi) u^m, \tag{2}$$

where for $\chi \neq 1$ the L-function $L(u, \chi) = \sum_F \chi(F) u^{\deg F}$ is a polynomial of degree $\leq l - 1$. Its coefficients are the Fourier transforms on G_l of the class distributions of the monic polynomials of degree $< l$; $\psi_n(\chi)$ follows by the Newton recursion, and one inverse FFT on G_l returns $\Psi(x)$

TABLE 1. The zero blocks $Z_d = \{t^{2d} + s : \deg s < d\} \subset \mathbb{F}_2[t]$: squarefree members, members with disc $\equiv 5 \pmod 8$ (Frobenius odd; integer lift with coefficients $0, 1$), and irreducibles. Stickelberger–Swan holds on every squarefree member (script `swan_parity.py`).

d	$\lvert Z_d\rvert$	squarefree	disc $\equiv 5$	irreducible
2	4	2	2	1
3	8	4	3	1
4	16	8	2	**0**
5	32	16	10	4
6	64	32	20	6
7	128	64	42	9
8	256	128	64	10
9	512	256	130	28
10	1024	512	280	49

for all q^l classes at once. This is the algorithm of [3], run here at level $l = d$ and degree $n = 2d$. The number of irreducibles in x is

$$\pi(x) = \frac{1}{n}\Big(\Psi(x) - \sum_{k\geq 2,\ k\mid n} \sum_{\deg \varpi = n/k,\ \varpi^k \in x} \frac{n}{k}\Big),$$

where the proper prime powers are enumerated directly (Ben-Or test in degree $\leq d$) and placed by the homomorphism $\mathrm{class}(\varpi^k) = k \cdot \mathrm{class}(\varpi)$ in G_l. Each run checks that every $\pi(x)$ is an integer and that $\sum_x (n\pi(x) + E(x)) = q^n$, and reports the largest distance of the floating-point $\Psi(x)$ from an integer (below 10^{-7} in all runs). As an independent check, a brute-force census that tests every polynomial of degree $2d$ for irreducibility (`verify_pointwise_oppermann.py`) agrees with the FFT counts class by class for all $q^{2d} \leq 2^{22}$ (`compare_bruteforce.py`), and in minimum, mean and empty classes for all $q^{2d} \leq 2^{26}$.

4.2. **Results.** Table 2 lists, for odd q, the minimum of $\pi(x)$ over all classes (all of which are Oppermann sub-intervals by Lemma 2.1) and its ratio to the expected value $q^d/2d$. Table 3 gives the same for the Oppermann classes in characteristic 2. The full output, all degrees, is in the file
`data/hayes_oppermann_census.txt`.

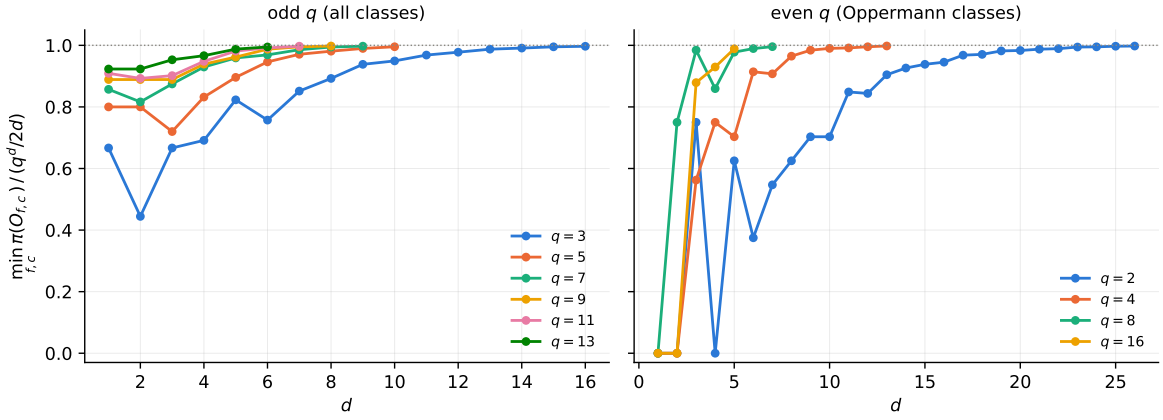


FIGURE 1. Normalised minimum $\min_{f,c} \pi(O_{f,c})/(q^d/2d)$ against d. Left: odd q. Right: even q; the points at 0 are the empty sub-intervals of Theorem 1.1.

TABLE 2. Odd q: irreducible counts π over all q^d Hayes classes of level d in degree $2d$ (all are Oppermann sub-intervals). Last four degrees for each q; all smaller degrees are in the data file, with minimum ≥ 1 throughout.

q	d	classes	$\min \pi$	mean π	$\min \pi/(q^d/2d)$
3	13	1 594 323	60 556	61 320.1	0.9875
	14	4 782 969	169 300	170 820.3	0.9911
	15	14 348 907	476 000	478 296.9	0.9952
	16	43 046 721	1 341 030	1 345 210.0	0.9969
5	7	78 125	5 418	5 580.3	0.9709
	8	390 625	23 951	24 414.0	0.9810
	9	1 953 125	107 410	108 506.9	0.9899
	10	9 765 625	486 042	488 281.2	0.9954
7	6	117 649	9 500	9 804.0	0.9690
	7	823 543	57 981	58 824.4	0.9857
	8	5 764 801	358 343	360 300.0	0.9946
	9	40 353 607	2 235 915	2 241 867.0	0.9973
9	5	59 049	5 680	5 904.8	0.9619
	6	531 441	43 728	44 286.7	0.9874
	7	4 782 969	339 993	341 640.6	0.9952
	8	43 046 721	2 684 630	2 690 420.0	0.9978
11	4	14 641	1 735	1 830.0	0.9480
	5	161 051	15 804	16 105.0	0.9813
	6	1 771 561	146 403	147 630.0	0.9917
	7	19 487 171	1 388 036	1 391 940.7	0.9972
13	3	2 197	349	366.0	0.9531
	4	28 561	3 450	3 570.0	0.9664
	5	371 293	36 672	37 129.2	0.9877
	6	4 826 809	400 104	402 234.0	0.9947

Proof of Theorems 1.2 and 1.3. Exact computation as described above; the minima are in Tables 2 and 3, and the empty classes in characteristic 2 occur exactly at

$$(q, d) \in \{(2, 1), (2, 2), (2, 4), (4, 1), (4, 2), (8, 1), (16, 1), (16, 2)\},$$

one class each, which are the classes of Theorem 1.1. □

The normalised minimum $\min_x \pi(x)/(q^d/2d)$ tends to 1 in every column: the worst Oppermann sub-interval is only slightly below average, far from the Weil bound discussed next.

5. The square-root barrier

In (2) with $n = 2d$, $l = d$, the main term is q^d. By Weil, every inverse zero γ of $L(u, \chi)$, $\chi \neq 1$, has $|\gamma| \in \{1, \sqrt{q}\}$, so $|\psi_{2d}(\chi)| \leq (d-1)q^d$, and averaging the $q^d - 1$ nontrivial characters gives only

$$|\Psi(x) - q^d| \leq (d-1)q^d.$$

For $d \geq 2$ the error term is at least the main term, for every q. This is the exact analogue of the situation over $\mathbb{Z}$, where Oppermann's intervals have length $\sqrt{x}$ and the Riemann hypothesis falls short by $\log x$. So no degree threshold beyond which $O_{f,c} \neq \emptyset$ follows from Weil's theorem, and Theorem 1.1 shows that in characteristic 2 no such threshold can be uniform in q for $d = 2$.

Results on the variance of Ψ over Legendre intervals (an average over f) bound the number of intervals that deviate by a given amount; they do not exclude a single empty interval. For fixed degree and $q \to \infty$, the Galois-theoretic method of Bank, Bary-Soroker and Rosenzweig

TABLE 3. Even q: irreducible counts over the $q^{\lceil d/2 \rceil}$ Oppermann classes (Lemma 2.2). Last four degrees for each q; the only empty classes in the whole range are those of Theorem 1.1.

q	d	classes	$\min \pi$	mean π	$\min \pi/(q^d/2d)$
2	23	4 096	181 349	182 329.3	0.9945
	24	4 096	347 822	349 412.9	0.9951
	25	8 192	669 169	671 025.2	0.9971
	26	8 192	1 287 576	1 290 408.9	0.9977
4	10	1 024	51 908	52 350.0	0.9901
	11	4 096	189 054	190 603.5	0.9916
	12	4 096	695 868	698 820.5	0.9954
	13	16 384	2 575 566	2 580 963.0	0.9979
8	4	64	440	490.0	0.8594
	5	512	3 204	3 276.0	0.9778
	6	512	21 616	21 840.0	0.9895
	7	4 096	149 148	149 796.0	0.9957
16	2	16	0	60.0	0.0000
	3	256	600	680.0	0.8789
	4	256	7 616	8 160.0	0.9297
	5	4 096	103 680	104 856.0	0.9888

[1] gives asymptotics for primes in short intervals in a range of lengths; our blocks B_e have the form $\{P + s : \deg s \leq 1\}$ in degree 4, and Theorem 1.1(B) gives a family $q = 4^j \to \infty$ with zero primes, so any such asymptotic must exclude this length in characteristic 2. We refer to [1] for the precise range of validity.

6. Questions

(1) In characteristic 2, are (A)–(C) the only empty Oppermann sub-intervals for all q and d? For $d = 3$, Lemma 2.2 leaves q^2 classes; for $d = 4$, q^2 classes, among them the zero block Z_4, which is nonempty for $q = 4, 8, 16$.
(2) In odd characteristic, is every Oppermann sub-interval nonempty for all q and d? Lemma 2.1 turns this into a statement about all short intervals of length $q^{n/2}$.
(3) Outside the characteristic-2 cases above, is there a constant $c > 0$ with $\min_x \pi(x) \geq c\, q^d/2d$ for all large d?

Data and code

- `hayes_oppermann.py`: Theorems 1.2 and 1.3.
- `verify_pointwise_oppermann.py`: brute-force census, $q^{2d} \leq 2^{26}$.
- `compare_bruteforce.py`: class-by-class cross-check.
- `make_tables.py`, `make_figure.py`: Tables 2, 3 and Figure 1.
- `char2_d2_blocks.py`, `char2_d2_theorem.py`: Theorem 1.1(B).
- `swan_parity.py`: Theorem 1.1(C) and Table 1.
- `hayes_fft_fq.py`, `fqlib.py`, `small_q_gf.py`: taken unchanged from [3].

All scripts are in `code/`; their outputs are in `data/`.

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